Application of Transposition Ciphers in an Anti-Scraping Image Protection System Using Tile-Level Transposition

November 4, 2026

1 Introduction

1.1 Problem statement: Automated Image Scraping and Economic Threats

1.2 Objectives

2 Theoretical Background and Mathematical Formulations

+Reading the content on the `srt_and_grt_handwritten_notes.md` file

2.1 Terminologies and Mathematical Notations

2.2 Generalized Rectangular Transform (GRT)

2.3 The 4 Prerequisite Conditions for Invertibility

2.4 The 4-Stage Exact Coordinate Inversion (IGRT)

3 Algorithm Design and System Implementation

+ implementation walkthough on src/ folder

3.1 End-to-End Threat Model and Anti-Scraping Workflow

3.2 Server-Side Scrambling (Encryption) implementation in Python (`grt_server.py`)

3.3 Client-Side Reconstruction (Decryption) Implementation in JavaScript(`igrt_client.j`

4 Experimental Evaluation and Results

+ Evaludation result on bechmark-report.mf file

4.1 Experimental Setup and Benchmark Methodology

4.2 Computational Execution Latency (Overhead

4.3 Memory Footprint and Allocation Stability

5 Conclusion, Limitations, and Future Work

5.1 Summary of Achievements

5.2 Limitations and Future Work